DLR-TomoSAR — Physics-constrained learning

five loss terms · curriculum · the unrolled solver

July 2026

Why constrain the solution space

- The labels are classical fits: the network's ceiling is the label quality — and Act IV showed the network happily leans on statistical shortcuts.
- Physics terms restrict predictions to profiles **consistent with the SAR signal model**:
  - today — as a regularizer: physical quantities recomputed from the predicted profile must match those of the GT profile;
  - next — as self-supervision against the *measured* covariance itself, removing the label ceiling entirely.
- Five terms implemented and theory-tested; every one differentiable end-to-end through the Gaussian-mixture curve reconstruction.

The shared forward operator

All terms use the vertical wavenumber and the steering matrix on the elevation grid:

$$\boxed{k_z^{(i)} = \frac{4\pi\, b_i}{\lambda\, r_0\, s}, \qquad A_{in} = e^{j\, k_z^{(i)}\, \xi_n}}$$

- Geometry resolved from campaign track parameters — not nominal defaults: per-pass perpendicular baselines, look angle, slant range.
- **Per-pixel k_z field** for range/azimuth-varying geometry, built at preprocessing and threaded dataset $\rightarrow$ trainer $\rightarrow$ loss.
- Steering sign settled empirically ($+k_z$: median peak error 0.67 m, within one Rayleigh cell).

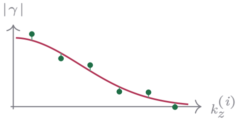
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n)\, e^{jk_z^{(i)}\xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i \left| \gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)}) \right|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences, baseline by baseline.

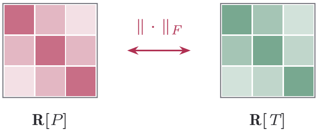
Cloude 2006 (PCT); same Fourier kernel as gamma-Net (Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A}\,\mathrm{diag}(P)\,\mathbf{A}^H \Delta\xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale, entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros & Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

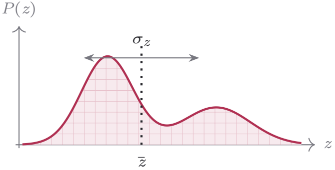
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n)\,\Delta\xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \qquad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



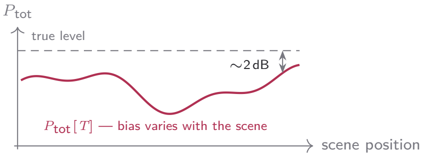
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\,\cdot\,] = \sum_n (\,\cdot\,)(\xi_n)\,\Delta\xi$$

$$\ell_{\text{pow}} = \left\langle \left(P_{\text{tot}}[P] - P_{\text{tot}}[T]\right)^2 \right\rangle$$

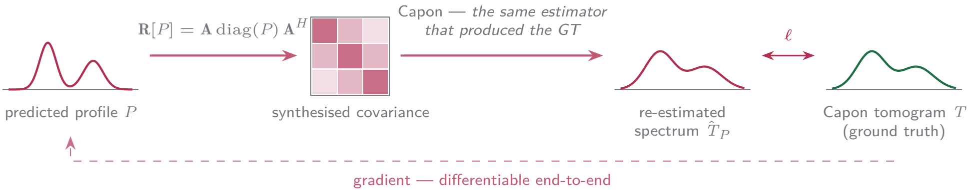


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

$$\hat{T}_P(\xi_n) = \left(\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n)\right)^{-1}$$



- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

Why the physics terms need a curriculum

The physics terms are the most sensitive consumers of the reconstructed profile — and at random initialisation they see only noise:

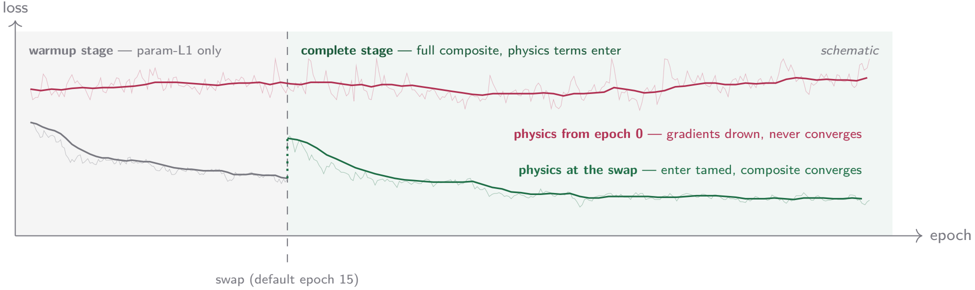
untrained network
predicted profiles are noise

→

physics forward operators
$\mathbf{R}[P]$, Capon solve —
ill-conditioned on unphysical profiles

→

huge, unstable losses
gradients drown the useful signal



Swap mechanics, calibration — and status

Each detail is an anticipated failure mode:

- **Resets at the swap**: LR schedule origin, LR warmup, Adam moments, early-stopping counters — optimizer state calibrated to the old loss is stale under the new one.
- **Checkpoint-baseline reset**: best-loss tracking restarts across the non-comparable loss scales — otherwise the pre-swap best could never be beaten.
- **Stage inheritance**: the complete stage is the base configuration, warmup a derived pre-phase materialised at launch — the stages cannot drift.
- **Loss-scale probe**: every active term run at unit weight over probe batches, IQR-filtered, folded into the per-term weights — physics terms enter at calibrated magnitude, not guessed weights.

Status

Implemented, SAR-theory-verified, geometry wired end-to-end — the single-term $K=2$ sweep is in (next frames); the Capon-cycle arm, the most expensive of the five, is still queued.

The physics sweep at $K=2$ — every term rides low, the scale terms collapse high

	reconstruction		detection (matched)						
objective	curve R^2 ↑	cos med ↑	prec. ↑	recall ↑	F1 ↑	exact ↑	under ↓	over ↓	peak p95 ↓
param-L1 + cos	0.747 ±.006	0.965 ±.004	0.895 ±.008	0.901 ±.001	0.898 ±.004	0.908 ±.007	0.041 ±.003	0.051 ±.009	4.7 ±.0
+ *coherence re-synthesis*									
$w=0.01$	0.752 ±.004	0.966 ±.002	0.895 ±.006	0.902 ±.001	0.898 ±.003	0.910 ±.002	0.040 ±.004	0.050 ±.005	4.7 ±.0
$w=0.05$	0.752 ±.006	0.966 ±.001	0.893 ±.005	0.902 ±.001	0.898 ±.002	0.911 ±.003	0.039 ±.003	0.051 ±.004	4.7 ±.0
$w=0.25$	0.755 ±.007	0.966 ±.002	0.892 ±.009	0.903 ±.002	0.897 ±.004	0.909 ±.005	0.038 ±.005	0.053 ±.010	4.7 ±.0
$w=0.5$	0.748 ±.009	0.965 ±.003	0.886 ±.013	0.903 ±.001	0.895 ±.007	0.911 ±.010	0.032 ±.006	0.057 ±.014	4.7 ±.0
$w=1$	0.751 ±.008	0.966 ±.002	0.870 ±.045	0.903 ±.001	0.886 ±.024	0.885 ±.063	**0.032** ±.005	0.083 ±.068	4.7 ±.0
+ *covariance matching*									
$w=0.01$	0.752 ±.001	0.965 ±.002	0.896 ±.005	0.901 ±.001	0.898 ±.003	0.906 ±.004	0.043 ±.001	0.051 ±.005	4.7 ±.0
$w=0.05$	0.748 ±.001	0.966 ±.002	0.899 ±.005	0.902 ±.001	0.900 ±.002	0.909 ±.004	0.043 ±.002	0.048 ±.005	4.7 ±.0
$w=0.25$	0.738 ±.004	0.966 ±.002	0.896 ±.006	0.899 ±.002	0.898 ±.003	0.902 ±.008	0.045 ±.002	0.053 ±.008	4.7 ±.0
$w=0.5$	0.727 ±.011	0.965 ±.002	0.900 ±.001	0.898 ±.002	0.899 ±.001	0.907 ±.003	0.046 ±.003	0.047 ±.001	4.7 ±.0
$w=1$	0.723 ±.006	0.965 ±.001	0.899 ±.005	0.899 ±.002	0.899 ±.002	0.903 ±.003	0.047 ±.004	0.050 ±.006	4.7 ±.0
+ *moments*									
$w=0.01$	0.753 ±.003	0.966 ±.002	0.895 ±.006	0.902 ±.001	0.898 ±.003	0.910 ±.004	0.040 ±.005	0.050 ±.006	4.7 ±.0
$w=0.05$	0.755 ±.002	0.966 ±.002	0.896 ±.005	0.902 ±.001	0.899 ±.002	0.910 ±.004	0.041 ±.004	0.049 ±.005	4.7 ±.0
$w=0.25$	0.751 ±.002	0.965 ±.002	0.897 ±.004	0.901 ±.001	0.899 ±.002	0.908 ±.003	0.044 ±.000	0.049 ±.004	4.7 ±.0
$w=0.5$	0.750 ±.004	0.966 ±.002	0.898 ±.003	0.900 ±.001	0.899 ±.001	0.909 ±.001	0.044 ±.002	0.047 ±.003	4.7 ±.0
$w=1$	0.746 ±.005	0.966 ±.001	0.897 ±.003	0.901 ±.001	0.899 ±.002	0.908 ±.003	0.043 ±.001	0.049 ±.003	4.7 ±.0
+ *total power*									
$w=0.01$	0.752 ±.003	0.965 ±.002	0.896 ±.005	0.902 ±.001	0.899 ±.002	0.910 ±.004	0.041 ±.004	0.049 ±.005	4.7 ±.0
$w=0.05$	0.751 ±.005	0.966 ±.001	0.897 ±.003	0.900 ±.001	0.899 ±.002	0.908 ±.002	0.044 ±.001	0.048 ±.002	4.7 ±.0
$w=0.25$	0.747 ±.003	0.965 ±.001	0.898 ±.003	0.900 ±.001	0.899 ±.001	0.908 ±.004	0.044 ±.003	0.049 ±.004	4.7 ±.0
$w=0.5$	0.737 ±.015	0.965 ±.002	0.900 ±.005	0.899 ±.002	0.899 ±.003	0.905 ±.006	0.048 ±.004	0.047 ±.006	4.7 ±.0
$w=1$	0.724 ±.020	0.963 ±.005	0.902 ±.003	0.898 ±.005	0.900 ±.001	0.907 ±.003	0.049 ±.007	**0.044** ±.004	4.8 ±.3

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks) · parity twins. Row = loss (param-L1 + 0.5 cos + w phys) / $(1.5 + w)$, one physics term per arm; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean ± seed std; held-out test, $5.0\,$M px. Validation losses are omitted — every arm optimizes a different composite, so they do not compare. **bold** = best in column, marked only where best and worst differ by more than 5%. Green = the best curve R^2 and the $w=1$ coherence under trade; red = its over price and the covariance / total-power collapse arms. cos med: median per-pixel cosine to the GT profile; peak p95: 95th percentile of the per-pixel peak error.

The $K{=}2$ physics detection by scatterer count — the set is untouched at every weight

objective	precision ↑ all	precision ↑ $k{=}1$	precision ↑ $k{=}2$	recall ↑ all	recall ↑ $k{=}1$	recall ↑ $k{=}2$	count acc | pred k ↑ $k{=}1$	count acc | pred k ↑ $k{=}2$
param-L1 + cos	0.895 ±.008	0.935 ±.010	0.828 ±.006	0.901 ±.001	0.999 ±.000	0.763 ±.002	0.944 ±.004	0.814 ±.024
+ *coherence re-synthesis*								
$w{=}0.01$	0.895 ±.006	0.935 ±.006	0.828 ±.006	0.902 ±.001	0.999 ±.000	0.765 ±.001	0.945 ±.005	0.815 ±.012
$w{=}0.05$	0.893 ±.005	0.935 ±.005	0.826 ±.005	0.902 ±.001	0.999 ±.000	0.765 ±.002	0.947 ±.004	0.814 ±.011
$w{=}0.25$	0.892 ±.009	0.932 ±.011	0.826 ±.006	0.903 ±.002	0.999 ±.000	0.766 ±.004	0.948 ±.006	0.808 ±.023
$w{=}0.5$	0.886 ±.013	0.928 ±.016	0.818 ±.010	0.903 ±.001	0.999 ±.000	0.767 ±.003	0.955 ±.008	0.803 ±.033
$w{=}1$	0.870 ±.045	0.902 ±.067	0.818 ±.009	0.903 ±.001	0.999 ±.000	0.768 ±.001	0.954 ±.002	0.755 ±.122
+ *covariance matching*								
$w{=}0.01$	0.896 ±.005	0.935 ±.006	0.831 ±.004	0.901 ±.001	0.999 ±.000	0.761 ±.002	0.943 ±.003	0.812 ±.016
$w{=}0.05$	0.899 ±.005	0.938 ±.006	0.834 ±.003	0.902 ±.001	0.999 ±.000	0.763 ±.003	0.943 ±.002	0.819 ±.015
$w{=}0.25$	0.896 ±.006	0.932 ±.009	0.835 ±.002	0.899 ±.002	0.999 ±.000	0.758 ±.005	0.944 ±.003	0.803 ±.023
$w{=}0.5$	0.900 ±.001	0.939 ±.002	0.836 ±.002	0.898 ±.002	0.999 ±.000	0.755 ±.005	0.943 ±.005	0.820 ±.005
$w{=}1$	0.899 ±.005	0.936 ±.007	0.837 ±.002	0.899 ±.002	0.999 ±.000	0.756 ±.005	0.942 ±.007	0.812 ±.015
+ *moments*								
$w{=}0.01$	0.895 ±.006	0.936 ±.007	0.827 ±.007	0.902 ±.001	0.999 ±.000	0.764 ±.002	0.945 ±.006	0.816 ±.015
$w{=}0.05$	0.896 ±.005	0.937 ±.006	0.828 ±.006	0.902 ±.001	0.999 ±.000	0.764 ±.003	0.944 ±.005	0.817 ±.015
$w{=}0.25$	0.897 ±.004	0.937 ±.004	0.831 ±.004	0.901 ±.001	0.999 ±.000	0.761 ±.003	0.941 ±.001	0.817 ±.011
$w{=}0.5$	0.898 ±.003	0.939 ±.004	0.832 ±.003	0.900 ±.001	0.999 ±.000	0.761 ±.003	0.942 ±.002	0.821 ±.009
$w{=}1$	0.897 ±.003	0.937 ±.004	0.832 ±.003	0.901 ±.001	0.999 ±.000	0.761 ±.002	0.943 ±.001	0.817 ±.009
+ *total power*								
$w{=}0.01$	0.896 ±.005	0.937 ±.006	0.828 ±.006	0.902 ±.001	0.999 ±.000	0.763 ±.001	0.944 ±.005	0.818 ±.013
$w{=}0.05$	0.897 ±.003	0.938 ±.003	0.831 ±.004	0.900 ±.001	0.999 ±.000	0.761 ±.003	0.941 ±.001	0.819 ±.007
$w{=}0.25$	0.898 ±.003	0.937 ±.004	0.833 ±.004	0.900 ±.001	0.999 ±.000	0.761 ±.003	0.942 ±.002	0.817 ±.011
$w{=}0.5$	0.900 ±.005	0.939 ±.007	0.834 ±.003	0.899 ±.002	0.999 ±.000	0.757 ±.006	0.937 ±.006	0.820 ±.018
$w{=}1$	0.902 ±.003	0.943 ±.005	0.834 ±.002	0.898 ±.005	0.999 ±.000	0.755 ±.011	0.934 ±.008	**0.829** ±.009

runs: dual UNet-skip trunks · set-pred · Hungarian · $K{=}2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks) · parity twins. All rows matched (Hungarian pairing); k = true scatterer count of the pixel. Five seeded replicas per arm, cells mean ± seed std; held-out test, $5.0\,$M px. **bold** = best in column, marked only where best and worst differ by more than 5%. Green = the high-weight coherence $k{=}2$ recall; red = its destabilised claimed pairs. Precision rewards under-prediction — rank on recall. count acc | pred k: share of pixels claiming exactly k scatterers whose true count is k.

The $K{=}2$ physics component errors — covariance buys position, coherence pays it

objective	μ MAE ↓ all	μ MAE ↓ $k{=}1$	μ MAE ↓ $k{=}2$	σ MAE ↓ all	σ MAE ↓ $k{=}1$	σ MAE ↓ $k{=}2$	a MAE ↓ all	a MAE ↓ $k{=}1$	a MAE ↓ $k{=}2$
param-L1 + cos	1.71 ±.07	0.36 ±.01	3.78 ±.14	0.774 ±.011	0.193 ±.002	1.668 ±.023	0.182 ±.004	0.058 ±.003	0.373 ±.006
+ *coherence re-synthesis*									
$w{=}0.01$	1.70 ±.06	0.35 ±.01	3.78 ±.12	0.770 ±.009	0.194 ±.003	1.654 ±.024	**0.180** ±.003	**0.057** ±.002	0.369 ±.004
$w{=}0.05$	1.72 ±.05	0.35 ±.01	3.82 ±.11	0.779 ±.014	0.195 ±.002	1.673 ±.035	0.180 ±.003	0.057 ±.003	**0.369** ±.004
$w{=}0.25$	1.72 ±.06	0.35 ±.00	3.82 ±.13	0.778 ±.011	0.192 ±.003	1.675 ±.016	0.182 ±.005	0.058 ±.004	0.373 ±.005
$w{=}0.5$	1.80 ±.10	0.35 ±.01	4.00 ±.22	0.804 ±.017	0.191 ±.002	1.730 ±.033	0.183 ±.004	0.058 ±.003	0.372 ±.004
$w{=}1$	1.81 ±.12	0.35 ±.00	4.02 ±.26	0.809 ±.020	0.194 ±.006	1.737 ±.035	0.185 ±.006	0.061 ±.008	0.373 ±.004
+ *covariance matching*									
$w{=}0.01$	1.66 ±.04	0.35 ±.01	3.70 ±.09	0.755 ±.014	0.192 ±.004	1.626 ±.032	0.182 ±.003	0.058 ±.003	0.374 ±.004
$w{=}0.05$	1.65 ±.02	0.35 ±.01	3.66 ±.05	0.754 ±.007	0.194 ±.002	1.620 ±.017	0.187 ±.005	0.061 ±.004	0.381 ±.007
$w{=}0.25$	1.62 ±.01	0.34 ±.01	3.62 ±.04	0.725 ±.011	0.186 ±.002	**1.566** ±.024	0.196 ±.004	0.065 ±.004	0.400 ±.006
$w{=}0.5$	**1.61** ±.01	0.34 ±.01	**3.61** ±.03	**0.724** ±.014	**0.186** ±.003	1.567 ±.035	0.205 ±.005	0.069 ±.004	0.418 ±.010
$w{=}1$	1.61 ±.02	0.35 ±.01	3.61 ±.04	0.728 ±.018	0.189 ±.008	1.575 ±.039	0.214 ±.012	0.071 ±.005	0.439 ±.024
+ *moments*									
$w{=}0.01$	1.71 ±.06	0.35 ±.01	3.79 ±.14	0.772 ±.017	0.195 ±.003	1.657 ±.041	0.180 ±.003	0.057 ±.002	0.369 ±.004
$w{=}0.05$	1.69 ±.04	0.35 ±.01	3.75 ±.09	0.771 ±.018	0.195 ±.003	1.657 ±.043	0.182 ±.003	0.058 ±.003	0.372 ±.002
$w{=}0.25$	1.68 ±.04	0.35 ±.01	3.72 ±.09	0.763 ±.005	0.195 ±.003	1.643 ±.015	0.186 ±.004	0.060 ±.003	0.380 ±.005
$w{=}0.5$	1.66 ±.03	0.35 ±.01	3.69 ±.07	0.759 ±.007	0.194 ±.002	1.634 ±.016	0.190 ±.004	0.062 ±.004	0.387 ±.005
$w{=}1$	1.66 ±.02	0.35 ±.01	3.69 ±.04	0.760 ±.008	0.196 ±.004	1.633 ±.012	0.192 ±.002	0.063 ±.003	0.391 ±.003
+ *total power*									
$w{=}0.01$	1.70 ±.05	0.35 ±.01	3.77 ±.10	0.771 ±.019	0.196 ±.004	1.656 ±.045	0.180 ±.003	0.057 ±.003	0.370 ±.003
$w{=}0.05$	1.67 ±.03	0.35 ±.00	3.72 ±.08	0.761 ±.005	0.194 ±.003	1.639 ±.015	0.184 ±.004	0.059 ±.004	0.377 ±.004
$w{=}0.25$	1.65 ±.03	0.35 ±.01	3.66 ±.05	0.753 ±.011	0.195 ±.004	1.619 ±.016	0.188 ±.005	0.060 ±.005	0.385 ±.006
$w{=}0.5$	1.63 ±.02	0.35 ±.01	3.62 ±.04	0.762 ±.022	0.197 ±.006	1.646 ±.052	0.192 ±.004	0.062 ±.003	0.396 ±.007
$w{=}1$	1.63 ±.02	0.35 ±.01	3.63 ±.03	0.777 ±.021	0.205 ±.008	1.672 ±.055	0.198 ±.005	0.063 ±.003	0.410 ±.012

runs: dual UNet-skip trunks · set-pred · Hungarian · $K{=}2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks) · parity twins. Component errors over matched pairs only (Hungarian pairing); k = true scatterer count of the pixel. Five seeded replicas per arm, cells mean ± seed std; held-out test, $5.0\,$M px. **bold** = best in column, marked only where best and worst differ by more than 5%. Green = the covariance $w{=}0.05$ μ; red = the full-weight coherence μ/σ price. μ, σ in elevation units; a in linear amplitude.

The $K{=}2$ physics curve fidelity — flat everywhere except the collapse arms

	curve			SSIM vs GT			peak
objective	MAE ↓	RMSE ↓	PSNR [dB] ↑	elev ↑	range ↑	azim ↑	mean ↓
param-L1 + cos	0.0370 $\pm$.0005	0.217 $\pm$.002	52.63 $\pm$.10	0.9671 $\pm$.0004	0.9510 $\pm$.0008	0.9610 $\pm$.0006	1.23 $\pm$.02
+ *coherence re-synthesis*							
$w{=}0.01$	0.0367 $\pm$.0003	0.215 $\pm$.002	52.71 $\pm$.07	0.9673 $\pm$.0002	0.9513 $\pm$.0006	0.9613 $\pm$.0004	1.22 $\pm$.01
$w{=}0.05$	0.0368 $\pm$.0003	0.215 $\pm$.003	52.72 $\pm$.10	0.9673 $\pm$.0002	0.9512 $\pm$.0005	0.9612 $\pm$.0004	1.22 $\pm$.01
$w{=}0.25$	0.0366 $\pm$.0003	**0.213** $\pm$.003	52.78 $\pm$.13	0.9674 $\pm$.0002	0.9516 $\pm$.0004	0.9615 $\pm$.0003	1.22 $\pm$.01
$w{=}0.5$	0.0370 $\pm$.0006	0.217 $\pm$.004	52.65 $\pm$.16	0.9670 $\pm$.0006	0.9504 $\pm$.0015	0.9607 $\pm$.0010	1.23 $\pm$.02
$w{=}1$	0.0371 $\pm$.0007	0.215 $\pm$.003	52.70 $\pm$.14	0.9670 $\pm$.0006	0.9499 $\pm$.0027	0.9606 $\pm$.0014	1.22 $\pm$.01
+ *covariance matching*							
$w{=}0.01$	0.0366 $\pm$.0001	0.215 $\pm$.001	52.72 $\pm$.02	0.9675 $\pm$.0002	0.9520 $\pm$.0004	0.9615 $\pm$.0003	1.22 $\pm$.01
$w{=}0.05$	0.0365 $\pm$.0002	0.217 $\pm$.000	52.65 $\pm$.02	0.9675 $\pm$.0002	0.9521 $\pm$.0004	0.9616 $\pm$.0002	1.21 $\pm$.01
$w{=}0.25$	0.0368 $\pm$.0002	0.221 $\pm$.002	52.49 $\pm$.06	0.9673 $\pm$.0001	0.9516 $\pm$.0003	0.9612 $\pm$.0002	1.23 $\pm$.01
$w{=}0.5$	0.0375 $\pm$.0007	0.225 $\pm$.005	52.31 $\pm$.18	0.9669 $\pm$.0005	0.9509 $\pm$.0008	0.9606 $\pm$.0007	1.23 $\pm$.01
$w{=}1$	0.0377 $\pm$.0005	0.227 $\pm$.002	52.24 $\pm$.09	0.9670 $\pm$.0003	0.9510 $\pm$.0006	0.9605 $\pm$.0005	1.24 $\pm$.01
+ *moments*							
$w{=}0.01$	0.0366 $\pm$.0002	0.215 $\pm$.001	52.73 $\pm$.05	0.9674 $\pm$.0002	0.9516 $\pm$.0006	0.9614 $\pm$.0004	1.22 $\pm$.01
$w{=}0.05$	0.0365 $\pm$.0002	0.214 $\pm$.001	52.76 $\pm$.04	0.9674 $\pm$.0002	0.9519 $\pm$.0004	0.9615 $\pm$.0003	1.22 $\pm$.01
$w{=}0.25$	0.0366 $\pm$.0002	0.215 $\pm$.001	52.71 $\pm$.04	0.9674 $\pm$.0002	0.9519 $\pm$.0004	0.9615 $\pm$.0004	1.22 $\pm$.01
$w{=}0.5$	0.0365 $\pm$.0003	0.216 $\pm$.002	52.69 $\pm$.07	0.9675 $\pm$.0002	0.9520 $\pm$.0005	0.9616 $\pm$.0004	1.22 $\pm$.01
$w{=}1$	0.0367 $\pm$.0003	0.218 $\pm$.002	52.61 $\pm$.08	0.9675 $\pm$.0003	0.9519 $\pm$.0004	0.9614 $\pm$.0003	1.22 $\pm$.01
+ *total power*							
$w{=}0.01$	0.0367 $\pm$.0002	0.215 $\pm$.001	52.71 $\pm$.06	0.9673 $\pm$.0002	0.9517 $\pm$.0005	0.9614 $\pm$.0004	1.22 $\pm$.00
$w{=}0.05$	0.0366 $\pm$.0001	0.215 $\pm$.002	52.70 $\pm$.08	0.9674 $\pm$.0001	0.9519 $\pm$.0004	0.9615 $\pm$.0003	1.22 $\pm$.01
$w{=}0.25$	0.0367 $\pm$.0003	0.217 $\pm$.001	52.64 $\pm$.05	0.9674 $\pm$.0002	0.9519 $\pm$.0005	0.9614 $\pm$.0004	1.23 $\pm$.01
$w{=}0.5$	0.0373 $\pm$.0011	0.221 $\pm$.006	52.48 $\pm$.24	0.9669 $\pm$.0009	0.9512 $\pm$.0013	0.9608 $\pm$.0011	1.22 $\pm$.01
$w{=}1$	0.0377 $\pm$.0015	0.227 $\pm$.008	52.27 $\pm$.31	0.9667 $\pm$.0011	0.9508 $\pm$.0016	0.9605 $\pm$.0014	1.23 $\pm$.04

runs: dual UNet-skip trunks · set-pred · Hungarian · $K{=}2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, $5.0\,$M px. **bold** = best in column, marked only where best and worst differ by more than 5%. Red = the collapse-arm RMSE. peak in elevation units; peak median (0.67) and p95 (4.7) are identical in every arm and are not tabulated.

The $K=2$ physics per-pixel tail — descriptor and covariance terms cap the blow-ups

	per-pixel R^2 ↑					per-pixel cosine ↑		
objective	med	mean	p_5	p_1	min	med	mean	p_1
param-L1 + cos	0.888 ±.007	−1.2 ±.8	+0.3 ±.1	−14.9 ±11.4	−13.2k ±3.3k	0.965 ±.004	0.934 ±.003	0.588 ±.011
+ *coherence re-synthesis*								
w=0.01	0.889 ±.005	−1.0 ±.8	+0.3 ±.1	−13.5 ±9.2	−12.7k ±7.4k	0.966 ±.002	0.935 ±.002	0.590 ±.006
w=0.05	0.888 ±.004	−1.2 ±.9	+0.3 ±.1	−14.8 ±8.1	−11.9k ±8.0k	0.966 ±.001	0.935 ±.002	0.587 ±.008
w=0.25	0.891 ±.006	−0.9 ±.5	+0.3 ±.0	−12.4 ±4.0	−12.2k ±4.8k	0.966 ±.002	0.936 ±.001	0.599 ±.009
w=0.5	0.888 ±.010	−9.6 ±12.0	+0.2 ±.2	−70.2 ±80.2	−66.3k ±77.4k	0.965 ±.003	0.935 ±.002	0.598 ±.006
w=1	0.870 ±.049	−3.7 ±4.3	−0.9 ±2.6	−50.2 ±71.0	−22.5k ±11.5k	0.966 ±.002	0.936 ±.001	**0.601** ±.007
+ *covariance matching*								
w=0.01	0.889 ±.008	+0.7 ±.0	+0.4 ±.0	−0.7 ±.2	−4.3k ±0.2k	0.965 ±.002	0.934 ±.001	0.575 ±.013
w=0.05	0.889 ±.009	+0.8 ±.0	+0.4 ±.0	−0.5 ±.1	−4.2k ±0.2k	0.966 ±.002	0.934 ±.001	0.574 ±.007
w=0.25	0.884 ±.006	+0.8 ±.0	+0.4 ±.0	−0.4 ±.0	−4.6k ±0.1k	0.966 ±.002	0.932 ±.001	0.544 ±.014
w=0.5	0.880 ±.007	+0.8 ±.0	+0.4 ±.0	−0.4 ±.0	−4.6k ±0.0k	0.965 ±.002	0.931 ±.001	0.529 ±.011
w=1	0.882 ±.004	+0.8 ±.0	+0.4 ±.0	**−0.3** ±.0	−4.6k ±0.1k	0.965 ±.001	0.929 ±.001	0.510 ±.025
+ *moments*								
w=0.01	0.888 ±.008	−0.1 ±.5	+0.3 ±.1	−7.1 ±6.6	−6.2k ±2.2k	0.966 ±.002	0.935 ±.001	0.588 ±.004
w=0.05	0.888 ±.009	+0.5 ±.2	+0.4 ±.0	−2.2 ±1.1	−4.4k ±1.2k	0.966 ±.002	0.935 ±.001	0.584 ±.006
w=0.25	0.888 ±.008	+0.7 ±.0	+0.4 ±.0	−0.8 ±.2	−4.2k ±0.1k	0.965 ±.002	0.934 ±.001	0.578 ±.004
w=0.5	0.887 ±.007	+0.8 ±.0	+0.4 ±.0	−0.5 ±.0	−4.3k ±0.3k	0.966 ±.002	0.934 ±.001	0.570 ±.009
w=1	0.886 ±.006	**+0.8** ±.0	**+0.4** ±.0	−0.4 ±.0	−4.4k ±0.3k	0.966 ±.001	0.933 ±.001	0.562 ±.008
+ *total power*								
w=0.01	0.889 ±.009	+0.5 ±.2	+0.3 ±.0	−2.6 ±1.2	**−3.6k** ±0.2k	0.965 ±.002	0.935 ±.002	0.584 ±.005
w=0.05	0.889 ±.007	+0.7 ±.0	+0.4 ±.0	−0.9 ±.4	−4.1k ±0.1k	0.966 ±.001	0.934 ±.001	0.579 ±.008
w=0.25	0.888 ±.007	+0.8 ±.0	+0.4 ±.0	−0.5 ±.3	−4.3k ±0.3k	0.965 ±.001	0.933 ±.001	0.563 ±.016
w=0.5	0.887 ±.009	+0.8 ±.0	+0.4 ±.0	−0.5 ±.3	−4.2k ±0.3k	0.965 ±.002	0.931 ±.002	0.551 ±.010
w=1	0.885 ±.011	+0.7 ±.0	+0.4 ±.0	−0.8 ±.5	−4.2k ±0.2k	0.963 ±.005	0.929 ±.005	0.529 ±.026

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack → both trunks) · parity twins. Five seeded replicas per arm, cells mean ± seed std; held-out test, 5.0 M px; pixel R^2 min and its spread in thousands (k). **bold** = best in column, marked only where best and worst differ by more than 5%. Green = covariance capping the tail; red = the coherence w=1 tail. cos: per-pixel cosine to the GT profile.

Context: where does the physics live?

Three places the SAR signal model can enter a learned inversion — the project now covers all three:

in the labels
model zoo backbones (Act V)
classical fits supervise a generic network — it learns the inversion from scratch; 10^5–10^7 params

→

in the loss
physics-constrained terms (Act VI)
predictions must re-synthesise coherences and covariances; the network is unchanged

→

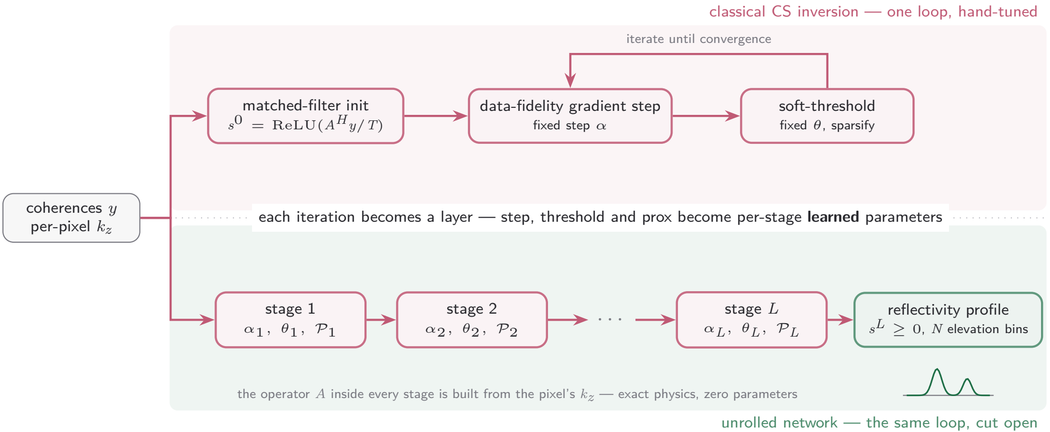
in the architecture
the unrolled solver (next)
the exact operator inside every layer; learned: $\alpha_\ell, \theta_\ell, \mathcal{P}_\ell$ — $\sim$1.4k params

learned from data

fixed physics

The unrolled family probes the far end of the axis — **the question it answers**: how much of the zoo's capacity is spent re-learning physics we already know?

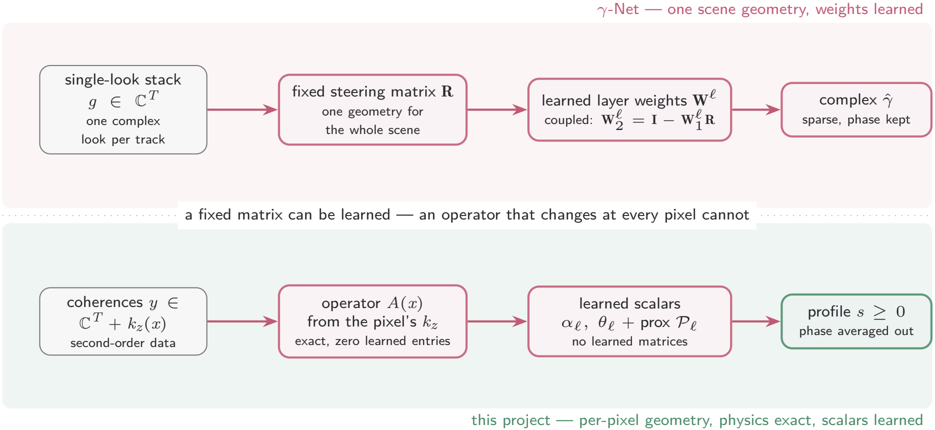
The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \,\mathrm{Re}\big[A^H(y - As^\ell)\big] / L_{\mathrm{lip}}, \qquad s^{\ell+1} = \mathrm{ReLU}\big(\mathcal{P}_\ell(r^\ell) - \theta_\ell\big)$$

γ-Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\mathrm{lip}} = TN\,\mathrm{d}z^2$ normalises the step. Default $L{=}8$: $\sim$1.4k parameters.

Adapting γ-Net — a different unknown, a per-pixel operator



γ-Net inverts $\quad g = \mathbf{R}\,\gamma + \varepsilon, \quad \gamma \in \mathbb{C}^N$ $\qquad$ this project inverts $\quad y = A(x)\,s, \quad s \in \mathbb{R}^N_{\geq 0}$

g: one complex look per track (first-order); y: per-track coherences (second-order) — averaging interferograms cancels the speckle phase, so the recoverable object is the power profile s, not the complex reflectivity γ. ε: measurement noise; T: tracks; N: elevation bins; $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation.

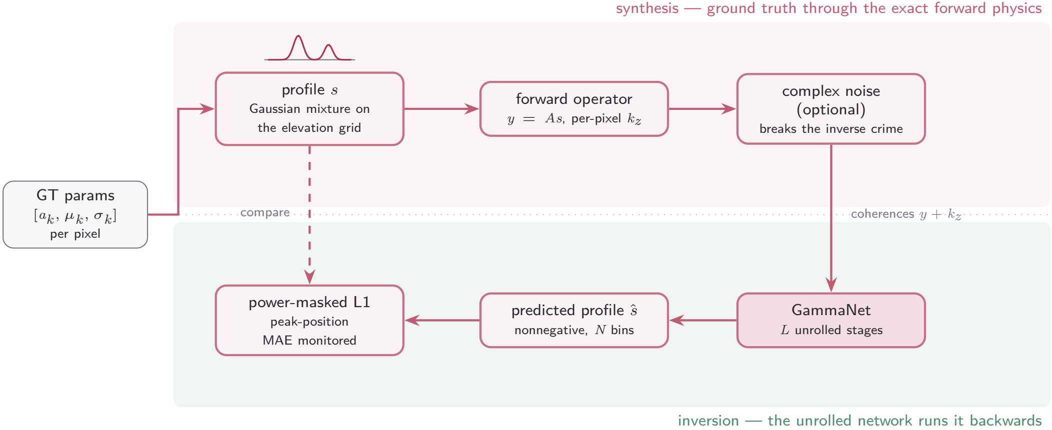
Adapting γ-Net — mechanism by mechanism

mechanism	γ-Net (Qian et al. 2022)		this project	why
iteration	gradient step $\to$ shrinkage, L stages	$=$	kept	the unrolling premise itself
supervision	train on simulated measurements	$=$	kept	the exact operator generates the pairs
unknown	complex reflectivity $\gamma \in \mathbb{C}^N$	$\to$	**power profile** $s \in \mathbb{R}^N_{\geq 0}$	speckle phase averages out of coherences
measurements	single-look stack g	$\to$	**per-track coherences** y	the pipeline is second-order end to end
operator	one fixed $\mathbf{R}$, scene-wide	$\to$	$A(x)$ **from the pixel's** k_z	airborne geometry varies across the scene
learned weights	$\mathbf{W}^\ell$ per stage, coupled to $\mathbf{R}$	$\to$	**none — scalars** $\alpha_\ell,\ \theta_\ell$	a per-pixel matrix cannot be learned
shrinkage	support selection + piecewise-linear function	$\to$	**prox** $\mathcal{P}_\ell$, **then** $\mathrm{ReLU}(\cdot - \theta_\ell)$	the prior must respect $s \geq 0$
model order	Bayesian information criterion after inversion	$\to$	**downstream parameter extraction**	the pipeline already fits K scatterers

The contract every change preserves: **the physics stays exact inside the network — only the prior is learned.**

support selection: the largest-magnitude entries bypass the threshold at each stage; piecewise-linear function: γ-Net's learned replacement for soft-thresholding; coupled weights: $\mathbf{W}_2^\ell = \mathbf{I} - \mathbf{W}_1^\ell \mathbf{R}$ halves the free parameters per stage. γ-Net reaches its accuracy at 12 stages; default here $L = 8$. green = exact physics carried by the operator · red = learned · grey = kept from γ-Net.

Training the unrolled model against the exact operator



Phase 1 (implemented): learn to invert the *exact* operator on synthesised coherences. Phase 2 (pending): measured covariances from the stack — no label ceiling.

Isolated family — own registry, trainer and entry point (`train_unrolled`); optimizer splits steps/thresholds from the prox; no AMP, the operator is complex-valued.

Why it exists: the physics-native comparison point — if ∼1.4k parameters of encoded physics match capacity-matched vision backbones, those backbones are mostly re-learning known physics.